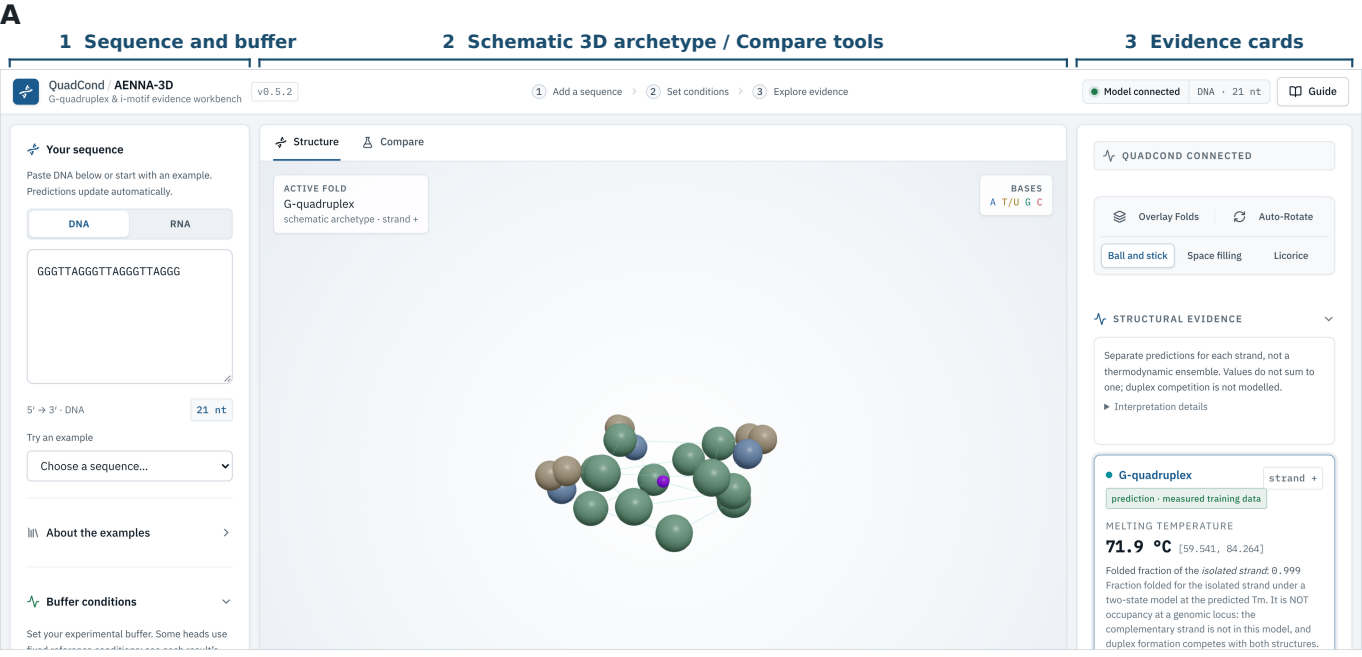


B Telomeric G4 · 140 mM K⁺, 10 mM Na⁺

G-quadruplex `strand +`

prediction · measured training data

MELTING TEMPERATURE

71.9 °C [59.541, 84.264]

Folded fraction of the *isolated strand*: 0.999
Fraction folded for the isolated strand under a two-state model at the predicted Tm. It is NOT occupancy at a genomic locus: the complementary strand is not in this model, and duplex formation competes with both structures.

> GGGTTAGGGTTAGGGTTAGGG
> head:g4_tm

C Same strand · no K⁺, 150 mM Na⁺

G-quadruplex `strand +`

prediction · measured training data

MELTING TEMPERATURE

58.7 °C [46.348, 71.071]

Folded fraction of the *isolated strand*: 0.992
Fraction folded for the isolated strand under a two-state model at the predicted Tm. It is NOT occupancy at a genomic locus: the complementary strand is not in this model, and duplex formation competes with both structures.

> GGGTTAGGGTTAGGGTTAGGG
> head:g4_tm

D Telomeric C-strand · reference buffer

i-motif `strand +`

prediction · measured training data

TRANSITIONAL PH

pH 6.38 [5.745, 7.009]

Folded fraction of the *isolated strand*: 0.027
Fraction folded for the isolated strand at your pH, from a Hill titration centred on the predicted transitional pH (cooperativity n = 2.5, an assumption, not fitted per sequence). Same caveat as the G4 card: not occupancy, and the duplex is not in the model.

> CCCTAACCCTAACCCTAACCCT
> head:im_pht

E C-strand in cytosol-like buffer: declined

⚠ 1 structure is hidden because the query falls outside what the head was trained on.

· k: every training record used k=100, so this head has no k response to apply to your k=140. Not an extrapolation -- an absence.

· mg: every training record used mg=0, so this head has no mg response to apply to your mg=1. Not an extrapolation -- an absence.

· temperature: every training record used temperature=25, so this head has no temperature response to apply to your temperature=37. Not an extrapolation -- an absence.

Use the i-motif model's buffer: K⁺ 100 mM · Mg²⁺ 0 mM · 25 °C

Show extrapolation anyway

F Compare › Conditions: K⁺ swept 0–150 mM, a verdict per head

Variant join | Mutations | Conditions | Batch | Model guide

Condition response

One axis, swept — with the heads that cannot answer named rather than hidden.

CSV JSON

K⁺ | Na⁺ | Mg²⁺ | pH

Sweep 0–150 mM

Swept sequence
GGGTTAGGGTTAGGGTTAGGG

Buffer
k 140 · na 10 · li_nh4 0 · mg 1 · ph 7.4 · temperature 37 · crowder_pct 0 · strand_conc 5

Model
0.4.6

Reverse complement (strand –)
CCCTAACCCTAACCCTAACCC

Draw extrapolated points. Off by default: a drawn line asserts a supported response between its endpoints, and these points are outside the head's domain and excluded from every summary below.

`g4_fold` strand + — too few supported points

0 of 12 points are inside this head's applicability domain, which is too few to describe a response. The extrapolated points are returned and drawn as gaps; they are deliberately not summarised.

12 of 12 points are outside this head's domain and are not in the summary above.

`g4_tm` strand + — responds

Across the supported range the prediction moves 33.5, against a typical error of 12.4. Descriptive, not a significance test.